HTTP: nginx и статический сайт

Конфигурация *nginx*:

```
admin@some:/etc/nginx/sites-available$ cat hw-hostadmin.local
server {
        listen 88;
        listen [::]:88;
        root /var/www/hw-hostadmin.local/html;

        index index.html;

        server_name _;

        location / {
                try_files $uri $uri/ =404;
        }
}
```

Символическая ссылка:

```
admin@some:/etc/nginx/sites-enabled$ ls -lah
total 8.0K
drwxr-xr-x 2 root root 4.0K Apr  6 23:27 .
drwxr-xr-x 8 root root 4.0K Apr  6 22:54 ..
lrwxrwxrwx 1 root root   34 Apr  5 17:29 default -> /etc/nginx/sites-available/default
lrwxrwxrwx 1 root root   45 Apr  6 23:27 hw-hostadmin.local -> /etc/nginx/sites-available/hw-hostadmin.local
```

html-файл:

```
admin@some:/var/www/hw-hostadmin.local/html$ cat index.html
<html>
<body>
    <p> Potapnev Maksim Andreevich </p>
    <p> Today: <span id="date"></span></p>
    <a href="http://hw-hostadmin.local:5000/">Redirect to Dotnet App (unit service)</a>


    <script>
        const now = new Date();
        const formattedDate = now.toLocaleDateString('ru-RU');
        document.getElementById('date').innerHTML = formattedDate;
    </script>
</body>
</html>
```

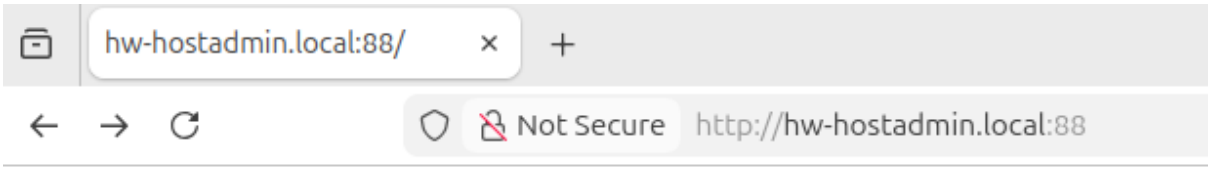
/etc/hosts:

```
admin@some:/etc/nginx/sites-enabled$ cat /etc/hosts
127.0.0.1       localhost.localdomain   localhost  hw-hostadmin.local
```

Результат вывода через *curl -v*:

```
admin@some:/etc/nginx/sites-enabled$ curl -v http://hw-hostadmin.local:88/
* Host hw-hostadmin.local:88 was resolved.
* IPv6: (none)
* IPv4: 127.0.0.1
*   Trying 127.0.0.1:88...
* Connected to hw-hostadmin.local (127.0.0.1) port 88
> GET / HTTP/1.1
> Host: hw-hostadmin.local:88
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Server: nginx/1.24.0 (Ubuntu)
< Date: Mon, 06 Apr 2026 21:35:29 GMT
< Content-Type: text/html
< Content-Length: 371
< Last-Modified: Mon, 06 Apr 2026 21:01:03 GMT
< Connection: keep-alive
< ETag: "69d41f0f-173"
< Accept-Ranges: bytes
<
<html>
<body>
    <p> Potapnev Maksim Andreevich </p>
    <p> Today: <span id="date"></span></p>
    <a href="http://hw-hostadmin.local:5000/">Redirect to Dotnet App (unit service)</a>


    <script>
        const now = new Date();
        const formattedDate = now.toLocaleDateString('ru-RU');
        document.getElementById('date').innerHTML = formattedDate;
    </script>
</body>
</html>
* Connection #0 to host hw-hostadmin.local left intact
```

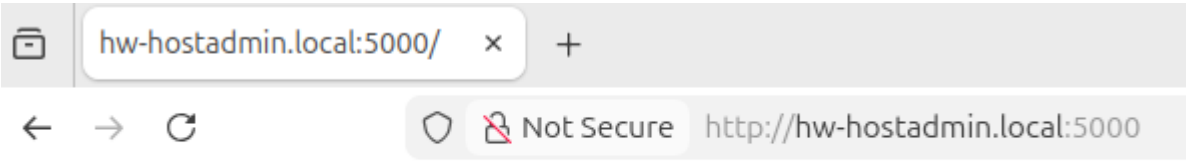
Результат из браузера + демонстрация связи с бэком:



Potapnev Maksim Andreevich

Today: 07.04.2026

Redirect to Dotnet App (unit service)



Hello from dotnet unit-service. Add /entities/(1..10) to path to get entity.

OpenSSL: просмотр “чужого” и своего сертификата

1. *Subject*(сервисы google)/*issuer*(центр сертификации для google и сторонних пользователей)/*notBefore*(валиден с 16 марта 08:36:41 GMT)/*notAfter*(валиден до 8 июня 08:36:40 GMT):

```
admin@some:/etc/nginx/sites-enabled$ openssl s_client -connect google.com:443 | openssl x509 -noout -subject -issuer -dates
depth=2 C = US, O = Google Trust Services LLC, CN = GTS Root R4
verify return:1
depth=1 C = US, O = Google Trust Services, CN = WE2
verify return:1
depth=0 CN = *.google.com
verify return:1
subject=CN = *.google.com
issuer=C = US, O = Google Trust Services, CN = WE2
notBefore=Mar 16 08:36:41 2026 GMT
notAfter=Jun  8 08:36:40 2026 GMT
```

2. В случае если на одном хосте развернуто несколько веб-серверов может быть несколько сертификатов как у главного хоста, так и у веб-серверов. Чтобы было понятно у какого именно запрашивать сертификат указывается *-servername.*

   SNI - это расширение протокола *TLS*, которое позволяет клиенту указывать имя хоста на начальном этапе рукопожатия. То есть *-servername* - это реализация возможности *SNI TLS*.

3. *SAN* для сертификата google:

```
            X509v3 Subject Alternative Name:
                DNS:*.google.com, DNS:*.appengine.google.com, DNS:*.bdn.dev, DNS:*.origin-test.bdn.dev, DNS:*.cloud.google.com, DNS:*.crowdsource.goog
le.com, DNS:*.datacompute.google.com, DNS:*.google.ca, DNS:*.google.cl, DNS:*.google.co.in, DNS:*.google.co.jp, DNS:*.google.co.uk, DNS:*.google.com.a
r, DNS:*.google.com.au, DNS:*.google.com.br, DNS:*.google.com.co, DNS:*.google.com.mx, DNS:*.google.com.tr, DNS:*.google.com.vn, DNS:*.google.de, DNS:
*.google.es, DNS:*.google.fr, DNS:*.google.hu, DNS:*.google.it, DNS:*.google.nl, DNS:*.google.pl, DNS:*.google.pt, DNS:*.googleapis.cn, DNS:*.gstatic.
cn, DNS:*.gstatic-cn.com, DNS:googlecnapps.cn, DNS:*.googlecnapps.cn, DNS:googleapps-cn.com, DNS:*.googleapps-cn.com, DNS:gkecnapps.cn, DNS:*.gkecnapp
s.cn, DNS:googledownloads.cn, DNS:*.googledownloads.cn, DNS:recaptcha.net.cn, DNS:*.recaptcha.net.cn, DNS:recaptcha-cn.net, DNS:*.recaptcha-cn.net, DN
S:widevine.cn, DNS:*.widevine.cn, DNS:ampproject.org.cn, DNS:*.ampproject.org.cn, DNS:ampproject.net.cn, DNS:*.ampproject.net.cn, DNS:google-analytics
-cn.com, DNS:*.google-analytics-cn.com, DNS:googleadservices-cn.com, DNS:*.googleadservices-cn.com, DNS:googlevads-cn.com, DNS:*.googlevads-cn.com, DN
S:googleapis-cn.com, DNS:*.googleapis-cn.com, DNS:googleoptimize-cn.com, DNS:*.googleoptimize-cn.com, DNS:doubleclick-cn.net, DNS:*.doubleclick-cn.net
, DNS:*.fls.doubleclick-cn.net, DNS:*.g.doubleclick-cn.net, DNS:doubleclick.cn, DNS:*.doubleclick.cn, DNS:*.fls.doubleclick.cn, DNS:*.g.doubleclick.cn
, DNS:dartsearch-cn.net, DNS:*.dartsearch-cn.net, DNS:googletraveladservices-cn.com, DNS:*.googletraveladservices-cn.com, DNS:googletagservices-cn.com
, DNS:*.googletagservices-cn.com, DNS:googletagmanager-cn.com, DNS:*.googletagmanager-cn.com, DNS:googlesyndication-cn.com, DNS:*.googlesyndication-cn
.com, DNS:*.safeframe.googlesyndication-cn.com, DNS:app-measurement-cn.com, DNS:*.app-measurement-cn.com, DNS:gvt1-cn.com, DNS:*.gvt1-cn.com, DNS:gvt2
-cn.com, DNS:*.gvt2-cn.com, DNS:2mdn-cn.net, DNS:*.2mdn-cn.net, DNS:googleflights-cn.net, DNS:*.googleflights-cn.net, DNS:admob-cn.com, DNS:*.admob-cn
.com, DNS:*.gemini.cloud.google.com, DNS:googlesandbox-cn.com, DNS:*.googlesandbox-cn.com, DNS:*.safenup.googlesandbox-cn.com, DNS:*.gstatic.com, DNS:
*.metric.gstatic.com, DNS:*.gvt1.com, DNS:*.gcpcdn.gvt1.com, DNS:*.gvt2.com, DNS:*.gcp.gvt2.com, DNS:*.url.google.com, DNS:*.youtube-nocookie.com, DNS
:*.ytimg.com, DNS:ai.android, DNS:android.com, DNS:*.android.com, DNS:*.flash.android.com, DNS:g.cn, DNS:*.g.cn, DNS:g.co, DNS:*.g.co, DNS:goo.gl, DNS
:www.goo.gl, DNS:google-analytics.com, DNS:*.google-analytics.com, DNS:google.com, DNS:googlecommerce.com, DNS:*.googlecommerce.com, DNS:ggpht.cn, DNS
:*.ggpht.cn, DNS:urchin.com, DNS:*.urchin.com, DNS:youtu.be, DNS:youtube.com, DNS:*.youtube.com, DNS:music.youtube.com, DNS:*.music.youtube.com, DNS:y
outubeeducation.com, DNS:*.youtubeeducation.com, DNS:youtubekids.com, DNS:*.youtubekids.com, DNS:yt.be, DNS:*.yt.be, DNS:android.clients.google.com, D
NS:*.android.google.cn, DNS:*.chrome.google.cn, DNS:*.developers.google.cn, DNS:*.aistudio.google.com
```

HTTPS на nginx (вариант 1)

Итоговая папка с сертификатами и ключами (*root.crt*/*root.key* - ключ и сертификат CA, *root.srl* - содержит серийные номера подписанных сертификатор, *san.cnf* - содержит возможные доменные имена, *server.crt* - серверный сертификат, *server.csr* - запрос на подпись сертификата, *server.key* - ключ сертификата сервера):

```
admin@some:~/ssl_hw/root$ ls
root.crt  root.key  root.srl  san.cnf  server.crt  server.csr  server.key
```

Конфигурация сервера:

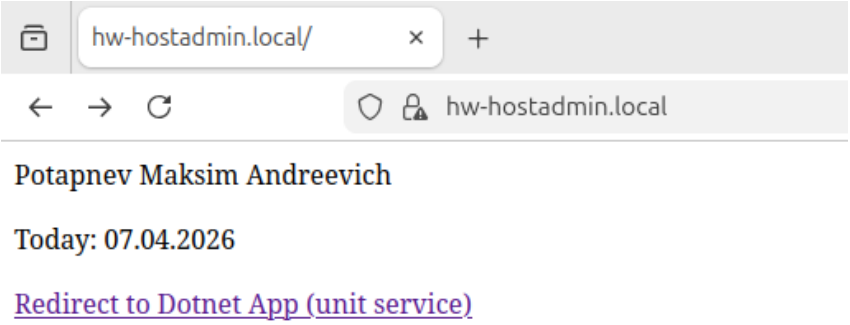
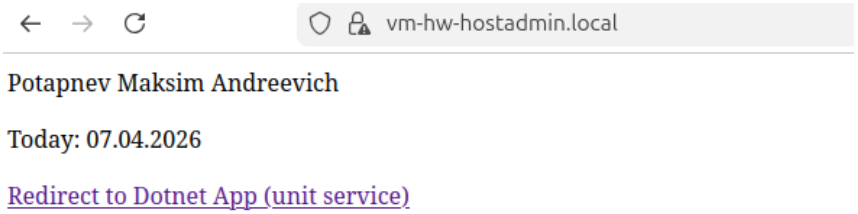
```
server {
        listen 88;
        listen [::]:88;

        return 301 https://$host$request_uri;
```

```
server {
        listen 443 ssl;
        listen [::]:443 ssl;
        root /var/www/hw-hostadmin.local/html;
        server_name hw-hostadmin.local vm-hw-hostadmin.local;

        index index.html;

        server_name _;

        ssl_certificate /home/admin/ssl_hw/root/server.crt;
        ssl_certificate_key /home/admin/ssl_hw/root/server.key;

        location / {
                try_files $uri $uri/ =404;
        }
}
```

Вывод в терминале:

```
admin@some:~/ssl_hw$ curl -v https://hw-hostadmin.local
* Host hw-hostadmin.local:443 was resolved.
* IPv6: (none)
* IPv4: 127.0.0.1
*   Trying 127.0.0.1:443...
* Connected to hw-hostadmin.local (127.0.0.1) port 443
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
*  CAfile: /etc/ssl/certs/ca-certificates.crt
*  CApath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / X25519 / RSASSA-PSS
* ALPN: server accepted http/1.1
* Server certificate:
*  subject: C=BY; ST=Minsk; L=Minsk; O=Maksim; CN=Maksim
*  start date: Apr  7 19:59:47 2026 GMT
*  expire date: Apr  7 19:59:47 2027 GMT
*  subjectAltName: host "hw-hostadmin.local" matched cert's "hw-hostadmin.local"
*  issuer: C=BY; ST=Minsk; L=Minsk; O=org; OU=Maksim; CN=Maksim; emailAddress=-
*  SSL certificate verify ok.
*   Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
*   Certificate level 1: Public key type RSA (4096/152 Bits/secBits), signed using sha256WithRSAEncryption
* using HTTP/1.x
> GET / HTTP/1.1
> Host: hw-hostadmin.local
> User-Agent: curl/8.5.0
```

Вывод в браузере:



Для второго домена также: